

Braille Input Device Reference Manual

1. System Overview

This document serves as a complete reference map for entering characters and commands using the USB Human Interface Device (HID) based on the CircuitPython firmware (code.py).

1.1 Keypad Mapping

The device uses a matrix keypad where keys are referenced by their (Row, Column) coordinate. Rows 3 and 4 are dedicated to Braille dot entry.

Braille Dot Number	Keypad Position (Row, Col)	Purpose
1	(3, 0)	Braille Dot Entry
2	(3, 1)	Braille Dot Entry
3	(3, 2)	Braille Dot Entry
4	(3, 3)	Braille Dot Entry
5	(4, 0)	Braille Dot Entry
6	(4, 1)	Braille Dot Entry
N/A	(1, 0)	CONFIRM / SPACE / MODE SELECT

1.2 Entry Rule

1. Press the required Braille dots simultaneously.
 2. Confirm the entry by pressing the CONFIRM key (1, 0) or by waiting 2.0 seconds for Auto-Confirmation.
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2. Braille Character Mappings

2.1 Alphabet (Lowercase)

This is the default entry mode.

Character	Braille Dot Combination	Character	Braille Dot Combination
a	Dot 1	n	Dots 1, 3, 4, 5
b	Dots 1, 2	o	Dots 1, 3, 5
c	Dots 1, 4	p	Dots 1, 2, 3, 4
d	Dots 1, 4, 5	q	Dots 1, 2, 3, 4, 5
e	Dots 1, 5	r	Dots 1, 2, 3, 5
f	Dots 1, 2, 4	s	Dots 2, 3, 4
g	Dots 1, 2, 4, 5	t	Dots 2, 3, 4, 5
h	Dots 1, 2, 5	u	Dots 1, 3, 6
i	Dots 2, 4	v	Dots 1, 2, 3, 6
j	Dots 2, 4, 5	w	Dots 2, 4, 5, 6
k	Dots 1, 3	x	Dots 1, 3, 4, 6
l	Dots 1, 2, 3	y	Dots 1, 3, 4, 5, 6
m	Dots 1, 3, 4	z	Dots 1, 3, 5, 6

2.2 Mode Prefixes

These prefix codes change the input mode. Sequence: Press Dot Combination right arrow Press CONFIRM (1, 0).

Mode	Function	Braille Dot Combination
CAPITAL PREFIX	Applies an uppercase shift to the <i>next</i> character only.	Dot 6
NUMBER PREFIX	Switches input to the numeric character set.	Dots 3, 4, 5, 6
SYMBOL PREFIX	Switches input to the symbol character set.	Dots 4, 5

2.3 Numbers (0-9)

Requires the NUMBER PREFIX to be set first. The mode reverts to Alphabet after entry.

Number	Braille Dot Combination	Number	Braille Dot Combination
1	Dot 1	6	Dots 1, 2, 4
2	Dots 1, 2	7	Dots 1, 2, 4, 5
3	Dots 1, 4	8	Dots 1, 2, 5
4	Dots 1, 4, 5	9	Dots 2, 4
5	Dots 1, 5	0	Dots 2, 4, 5

2.4 Symbols

Requires the SYMBOL PREFIX to be set first. The mode reverts to Alphabet after entry.

Symbol	Braille Dot Combination	Symbol	Braille Dot Combination
!	Dot 1	&	Dots 1, 2, 4, 5
@	Dots 1, 2	*	Dots 1, 2, 5
#	Dots 1, 4	(	Dots 2, 4
\$	Dots 1, 4, 5	)	Dots 2, 4, 5
%	Dots 1, 5	^	Dots 1, 2, 4

3. Dedicated Command Keys

These keys execute standard HID commands immediately upon press and will clear and submit any pending Braille buffer.

3.1 Standard Keyboard & Navigation

Keypad Position (Row, Col)	Function	HID Action
\$(1, 1)\$	Backspace	Keycode.BACKSPACE
\$(1, 2)\$	Enter	Keycode.ENTER
\$(1, 3)\$	Space	Keycode.SPACE
\$(2, 1)\$	Up Arrow	Keycode.UP_ARROW
\$(2, 3)\$	Down Arrow	Keycode.DOWN_ARROW
\$(2, 0)\$	Right Arrow	Keycode.RIGHT_ARROW
(2, 2)	Left Arrow	Keycode.LEFT_ARROW
(5, 0)	Buzzer Toggle	Toggles the device's audible beep function.

3.2 Mouse and Click Commands

Keypad Position (Row, Col)	Function	HID Action
(5, 1)	Left Click	Mouse.LEFT_BUTTON
(5, 2)	Right Click	Mouse.RIGHT_BUTTON
(5, 3)	Middle Click	Mouse.MIDDLE_BUTTON

3.3 Custom Application Launchers

These combinations send unique three-key shortcuts intended for custom application mapping on the host PC (e.g., using a program like PyKey).

Keypad Position (Row, Col)	Shortcut Keys
(0, 0)	CTRL + ALT + F13
(0, 1)	CTRL + ALT + F14
(0, 2)	CTRL + ALT + F15
(0, 3)	CTRL + ALT + F16
JOY_BTN (Joystick Click)	CTRL + ALT + F17